

Javascript

DOM, dom events

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>DOM</title>
  </head>
  <body>
    <form id="myform" action="./funcArrayObject.html">
      <input id="input" type="text" placeholder="Enter your name" />
      <button type="submit">Submit</button>
    </form>

    <p id="inner">Innterhtml</p>
    <h1>DOM -> Document object Model</h1>

    <h2>H2 Tag</h2>
    <h2>H2 Tag</h2>

    <h3 id="one">Id tag</h3>

    <p class="item"></p>
    <p class="item"></p>
    <p class="item"></p>

    <p class="item"></p>
    <p class="item"></p>
    <p class="item"></p>

    <p class="item"></p>
    <p class="item"></p>
    <p class="item"></p>

    <ul>
      <li>item 1</li>
      <li class="active">item 2</li>
      <li class="active">item 2</li>
      <li>item 3</li>
      <li>item 4</li>
      <li>item 5</li>
      <p>Hello there</p>
    </ul>

    <script>
```

```
document.getElementById("myform").onsubmit = function (){
    alert("Form submitted");
}

document.getElementById("input").oninput = function(){
    console.log("input value" , this.value);
}

// document.getElementById("inner").textContent = '<b>hello</b>';
document.getElementById("inner").innerHTML = "<b>hello</b>";

document.getElementById("inner").style.backgroundColor = "blue";
document.getElementById("inner").style.fontSize = "30px";

const button = document.createElement("button");
button.textContent = "Submit";
button.classList.add("clickme");

button.addEventListener("click", function () {
    alert("You clicked button");
});

button.addEventListener("mouseover", function () {
    button.style.background = "blue";
});
button.addEventListener("mouseout", function () {
    button.style.background = "";
});

// document.addEventListener("keydown", function (event) {
//     console.log(event.key);
// });

// document.addEventListener("keyup", function (event) {
//     console.log(event.key);
// });

console.log(button);

document.body.appendChild(button);

// button.remove();

let tagName = document.getElementsByTagName("h2");
```

```

    let idName = document.getElementById("one");

    let className = document.getElementsByClassName("item");

    let queryAll = document.querySelectorAll("ul > p");
    let query = document.querySelector("ul > p");

    // console.log(queryAll);
    // console.log(query.tagName);

    // console.log(typeof className);

    // className[0].innerText = "Paragraph 1";

    for (let i = 0; i < className.length; i++) {
        className[i].innerText = `Paragraph ${i + 1}`;
        console.log(className[i].innerText);
    }

    // console.log(className[0].innerText);

    console.log(idName);

    console.log(tagName[0].innerText);
    console.log(tagName[1]);
</script>
</body>
</html>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Practice</title>
    <style>
        div{
            width: 100px;
            height: 100px;
            /* border-radius: 50%;
            border: 2px solid black; */

            background-color: burlywood;
        }

        .hidden{
            display: none;

```

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        /* background-color: yellow; */
    }
</style>
</head>
<body>

    <div id="box">
        Click below button
    </div>

    <button id="button">Click me to hide div container</button>

    <script>
        const but = document.getElementById("button");
        const box = document.getElementById("box");

        but.addEventListener("click", function(){
            // box.classList.add("hidden");
            box.classList.toggle("hidden");

        })
    </script>

</body>
</html>

```

Theory Notes

1. DOM Basics

- **DOM (Document Object Model)** → tree structure representation of HTML.
- **Selectors:**
 - `getElementById()` → single element by ID.
 - `getElementsByClassName()` → collection of elements by class.
 - `getElementsByTagName()` → collection of elements by tag.
 - `querySelector()` → first match.
 - `querySelectorAll()` → all matches.

2. Content Manipulation

- `innerHTML` → parses HTML tags.
- `textContent` → plain text only.
- `style.property` → inline styling.

- `classList.add()`, `classList.remove()`, `classList.toggle()` → manage CSS classes.

3. Events

- `onclick`, `oninput`, `onsubmit` → inline event handlers.
- `addEventListener()` → modern way to attach multiple events.
- Common events: `click`, `mouseover`, `mouseout`, `keydown`, `keyup`.

Coding Round Questions

1. Write a program to change the text of a `<p>` tag when a button is clicked.
2. Create a form that alerts “Form submitted” when submitted.
3. Use `querySelectorAll()` to select all `<li>` elements and print their text.
4. Demonstrate the difference between `innerHTML` and `textContent`.
5. Add a button that changes background color on `mouseover` and resets on `mouseout`.
6. Toggle a `hidden` class on a `<div>` when a button is clicked.
7. Write a program to log every key pressed using `keydown` event.

Challenge

Problem: Todo List Application

Requirements:

- Create an input box and a button.
- When the button is clicked, add the input value as a new `<li>` in a list.
- Each `<li>` should have a “Delete” button to remove it.
- Use DOM methods (`createElement`, `appendChild`, `remove`) and event listeners.
- Bonus: Toggle a “completed” class when an item is clicked.